# SE 423 Mechatronics

# Laboratory Assignment #2

**(One Week Lab)**

**Introduction to Raspberry PI4 and**

**Introduction to TMS320F28379D GPIO Programming and Texas Instruments Code Composer Studio**

**TA Check-Offs in this document: 10 (marked "TA Check-Off" in the text and listed at the end)**

## A few questions to get your brain thinking in binary number representation

1. If there was such thing as a 24 bit signed integer, what would be the largest positive number it could represent and what is the smallest negative number it could represent?
2. Below are three (signed) `int16_t` integers represented in binary format. What are these numbers in decimal format? Use the Windows Calculator and put it in Programmer Mode and select the WORD size (which is 16 bits).
   i. `1101110000011011`
   ii. `0001111100110101`
   iii. `1000000010110011`
3. In question 2i, is bit 10 high 1 or low 0? Remember we start numbering with bit 0 as right most bit.
4. ☐ **TA Check-Off 1: Meaning of the if statement**
   **Explain to your TA** what this if statement is checking for (think in binary) (& is bitwise AND)

```
if ((myregister & 0x4) == 0x4) {

}
```

## Goals for this Lab Assignment

1. Build a simple LabVIEW program to communicate over Ethernet (TCPIP) to the robot's Raspberry Pi4 board.
2. Use CPU Timer to periodically perform desired procedures/code.
3. Work with port inputs and port outputs.
4. What to do with a compiler error.
5. Debugging your source code with Breakpoints and the Watch Window.

## LED's Default GPIO Assignments

| | |
|---|---|
| LED1 | GPIO22, Controlled with Registers GPADAT, GPASET, GPACLEAR and GPATOGGLE |
| LED2 | GPIO94, Controlled with Registers GPCDAT, GPCSET, GPCCLEAR and GPCTOGGLE |
| LED3 | GPIO95, Controlled with Registers GPCDAT, GPCSET, GPCCLEAR and GPCTOGGLE |
| LED4 | GPIO97, Controlled with Registers GPDDAT, GPDSET, GPDCLEAR and GPDTOGGLE |
| LED5 | GPIO111, Controlled with Registers GPDDAT, GPDSET, GPDCLEAR and GPDTOGGLE |

Table 1: GPIO pin and registers of each LED

## Push Button's Default GPIO Assignments

| | |
|---|---|
| PB1 | GPIO157, Read bit status with Register GPEDAT |
| PB2 | GPIO158, Read bit status with Register GPEDAT |
| PB3 | GPIO159, Read bit status with Register GPEDAT |
| PB4 | GPIO160, Read bit status with Register GPFDAT |

Table 2: GPIO pin and register of each push button

**GPIO Register Use when GPIO pin set as Output:** The GPIO Registers are 32 bit registers but we use unions and bitfields in the C/C++ programming language to control just one bit of the 32 bit register at a time. The "`.all`" part of the C/C++ union is the entire 32bit register. The "`.bit.GPIO11`" is just one bit in the 32 bit register. So these two lines of C code perform the same operation:

```
//You have to think a little with this code to know that bit 11 is
//being set.
GpioDataRegs.GPASET.all = 0x00000800;
//This line of code is somewhat easier to understand that we are
//setting the 11th bit.
GpioDataRegs.GPASET.bit.GPIO11 = 1;
```

| Register | Usage | Example |
|---|---|---|
| GP?DAT | GP?DAT.bit.GPIO? = 1, Sets that Pin High, 3.3V<br>GP?DAT.bit.GPIO? = 0, Sets that Pin Low, 0V/GND<br>**BUT FOR TECHNICAL REASONS DO NOT USE DAT FOR TURNING ON AND OFF LEDS JUST USE TO READ STATE OF PUSHBUTTONS BELOW** | |
| GP?SET | GP?SET.bit.GPIO? = 1, Sets that Pin High, 3.3V<br>GP?SET.bit.GPIO? = 0, Does Nothing | **GpioDataRegs.GPBSET.bit.GPIO37 = 1; Sets GPIO37 High/3.3V**<br>**GpioDataRegs.GPBSET.bit.GPIO37 = 0; Does Nothing** |
| GP?CLEAR | GP?CLEAR.bit.GPIO? = 1, Sets that Pin Low, 0V/GND<br>GP?CLEAR.bit.GPIO? = 0, Does Nothing | **GpioDataRegs.GPCCLEAR.bit.GPIO70 = 1; Sets GPIO70 Low/0V**<br>**GpioDataRegs.GPCCLEAR.bit.GPIO70 = 0; Does Nothing** |
| GP?TOGGLE | GP?TOGGLE.bit.GPIO? = 1, Sets Pin opposite of its current state.<br>GP?TOGGLE.bit.GPIO? = 0, Does Nothing | **GpioDataRegs.GPDTOGGLE.bit.GPIO98 = 1; was 3.3V then 0V or was 0V then 3.3V**<br>**GpioDataRegs.GPDTOGGLE.bit.GPIO98 = 0; Does Nothing** |

Table 3: GPIO registers of a pin set as output

**GPIO Register Use When GPIO Pin Set as Input:** Each GPIO pin, when setup as an input, has an internal pull-up resistor that can either enabled/connected or disabled/disconnected to that GPIO pin. With the passive push button on our breakout board, we will need to enable the pull-up resistor.

| Register | Usage | Example |
|---|---|---|
| GP?DAT | If GP?DAT.bit.GPIO? is equal to 1 then the Pin is High, 3.3V<br>If GP?DAT.bit.GPIO? is equal to 0 then the Pin is Low, 0V/GND | `if (GpioDataRegs.GPADAT.bit.GPIO19 == 1) {`<br>`//code that needs to run when input pin`<br>`//GPIO19 is High/3.3V`<br>`} else {`<br>`// code that needs to run when input ping`<br>`// GPIO19 is Low/0V`<br>`}` |

Table 4: GPIO register of a pin set as input

## Laboratory Exercises

## Exercise 1

First, to give you practice performing these steps if needed in the future, check if the class repo has any changes or additional files to merge into your repo already on the lab PC. Under Lab 1, find the Git help file titled “Using the SE423 Repository” and read and perform the steps of the section of the document titled “Course File Updates.” These steps will pull the latest updates from the class repository. Ask your TA for help if needed. *There probably will not be any updates but in the future if your instructor makes some additions you will need to perform these steps again.*

In this exercise 1 you will create a simple LABVIEW VI from scratch to communicate with your robot. Use the below steps along with the picture of the block diagram below to create this LabVIEW application.

1. Open a new LABVIEW project and then its VI file. Remember Ctrl-E switches you back and forth between the front panel view and the block diagram view.

2. Then in File Explorer find your repo's folder and explore into "`workspace\LabView`". Create a folder "Lab2" here. *In "`workspace\LabView`" is where I recommend you save all your LabView files. This way when you add your CCS projects to Github by running "`git add workspace`" you will also be adding your LabView files.*

3. Save this new project and VI file to your "`workspace\LabView\Lab2`" folder.

4. Use the below block diagram picture and the remaining instructions to build your LABVIEW program.

5. Add a 4 frame flat sequence structure to your block diagram. The flat sequence will allow for initializations, the main body of the program and termination.

6. In the second frame add a While Loop.

7. Add a Stop button in the front panel view but do not wire the Stop button in the block diagram yet. Change the Mechanical action of the Stop button to "Switch When Released." For now in the block diagram view, place the Stop button outside the While Loop.

8. I like using the Mechanical Action "Switch When Released" (or "Switch When Pressed") for the Stop button but it requires initialization and termination code to make sure it is in the state you wish when you get to your main code. To control the state of the Stop button you can use a local variable to change its state. In the first sequence frame add a local variable located under "Structures" items. Click on the "?" of the local variable and select "Stop" (or whatever you named your stop button.) Connect a "False" Boolean constant to the input of the local variable. This will force the state of the Stop button to be false at the beginning of the program. The while loop stops when the Stop button returns true. For completeness add a "stop" local variable in the third sequence frame and also set it to false there.

9. In below steps we are going to setup a TCPIP connection between the LABVIEW program and a Linux program running on the robot. A small delay is necessary at the beginning and end during the TCPIP server connection and termination. So in the first and third sequence frame add a "Wait (ms)" of 1 second (1000 ms) to make sure the first and third frames take at least 1 second.

10. We will finish up the initialization and termination. Add a "TCP Open Connection" found under "Data Communication→Protocols→TCP to the first sequence frame. Right click at TCP Open Connection's "address". "remote port" and "timeout (ms)" ports and select create constant. Set the timeout to 1000 ms, remote port to 10001 and address to "192.168.1.7?" where ? is dependent on the robot you are using.

11. In the fourth sequence frame add a "TCP Close Connection."

12. Now we will add the main code to our VI. The goal of the VI is to open a TCPIP connection to a Linux program running on the robot car. Our front panel will have a string control, a Send button and a string indicator. The user will type a string to send to the robot car in the string control and then press the send button to send the text to the robot car. Then the robot car will receive that data and echo it back to the LABVIEW program and it will be displayed in the string indicator. This will be repeated until the Stop button is pressed.

13. Add an OK button to the front panel and make its Mechanical action "Switch When Released." Also change the OK text to SEND, and change the button's name to something like "sendbutton." For now place this button outside of the while loop.

14. Inside the While loop add an Event Structure. Add three events to this structure. One (labeled event "0" in the block diagram) being the default Timeout event. Second (labeled "1") a Send button Mouse Down event. Third (labeled "2") a Stop button Value Change event.

15. In the Send button event first place a TCP Write block and a Concatenate String block. Wire the "Connection ID" output of the "TCP Open Connection" block to the "Connection ID" input of the TCP Write block. Wire the TCP Write block timeout to a constant setting it to 100 ms. Wire the output of the Concatenate String

block to the “data in” port of the TCP Write block. Finally create an Indicator to display the number of bytes written output port.

16. Drag down on the Concatenate String block so that three items will be concatenated together. Create a constant string for the first element, a control string for the second element (call it “Send String”) and a constant for the third element. The Linux program is waiting for the start character 0xFD to begin the string and the stop character 0xFF to end the string. To send these hexadecimal numbers as a character select the constant string and right click on it. Down towards the bottom of the displayed menu you should find and select **“ ’\’ Codes Display”**. Do this for the two constants and type in the first “`\FD`” and the second “`\FF`”.

17. That completes the sending of a string to the robot, now wire the code to receive the string sent back to LABVIEW. Place a “TCP Read” block after the TCP Write block. First wire the “Connection ID” output of the TCP Write block to the “Connection ID” input of the “TCP Read” block. Also while we are here, wire the “Connection ID” output of the “TCP Read” block to the “Connection ID” input of the “TCP Close Connection” block which is located in the fourth sequence panel.

18. Create constants for the “TCP Read” block setting the Timeout to 3000 ms and the bytes to read to 100. Also create an indicator in the front panel view and connect its input to the “data out” of the “TCP Read” block in the block diagram. Name this indicator “Received String.”

19. The final addition for the “TCP Read” block is to create a constant to connect to the “mode” input. Once again, this is easiest to do by right clicking on the “mode” input node and selecting “Create→Constant”. The data transfer method between the robot and LABVIEW is slightly different then the LABVIEW to robot transfer. For the LABVIEW to robot transfer we placed a start and stop character at the beginning and end of the string. The “TCP Read” block can be set to implement a different transfer waiting until the carriage return and line feed characters are seen in sequence. This sequence indicates the end of the string. To put the “TCP Read” block in this mode, click the down arrow of the “mode” constant and select “CRLF.”

20. So that the event structure works correctly with the Send button that was created in step 12, move the Send button’s icon inside the Send Button Mouse Down Event.

21. The Timeout and Stop events are simple. In both events pass the “Connection ID” output of the “TCP Open Connection” block through to the output on its way to the “TCP Close Connection” block. For the Timeout event connect a False Boolean constant to the Stop button port. Also in the top left of the event structure there is a small hour glass icon. Right click on the input of this hour glass icon and create a constant and set the Timeout to 50ms. In the Stop button event connect a True Boolean constant to the Stop button port. Also here in the Stop button event place the Stop button icon. Double check the below picture, but your LABVIEW program should be finished now.

22. Next step is to get the correct program running on your Raspberry PI4 board on your robot. We need a TCPIP server application to be running that is waiting for text from your LABVIEW program and then will receive that test and send it right back (echo) to your LABVIEW program. You will use a program on your lab PCs called “putty.” To run putty, open a Windows cmd prompt. At the command prompt type “`putty pi@192.168.1.7?`” filling in the ? with your robot’s IP. (pi is the username @ its ip address) A terminal window should display and request you type in the password. Type the password, “`f33dback5`”, and then you have a wireless terminal of your RPI4’s Linux operating system. Type **ls** (list) and see all the folders of the user **pi**’s home directory. The program we want to run is called LVecho and it is located in the folder lvecho. Change directory into the lvecho folder by typing “`cd lvecho`” (cd is change directory). Run the Linux application **LVecho** by typing “`./LVecho`”. The “`./`” tells Linux to look in the current directory for the application to execute. This application will be used to verify your LABVIEW GUI is working. It simply receives the messages you send from LABVIEW, prints them to the Linux console and echoes them back to your LABVIEW application.

23. Run your LABVIEW application, and type a message into your Send String control. Click on the **Send** button and your message should print at the Linux console and echo back to the Received String indicator.

**☐ TA Check-Off 2: LabVIEW talking to the Raspberry Pi**

Demonstrate your LabVIEW programming talking back and forth to your robot's Raspberry Pi board.

24. When you are done playing with this end, click the STOP button and close the LVecho application at that Linux console by typing Ctrl-C.

25. **Leave the Raspberry PI on for now and leave the terminal running that is connected to your PI. You will be told to Shutdown the PI at the end of the lab.**

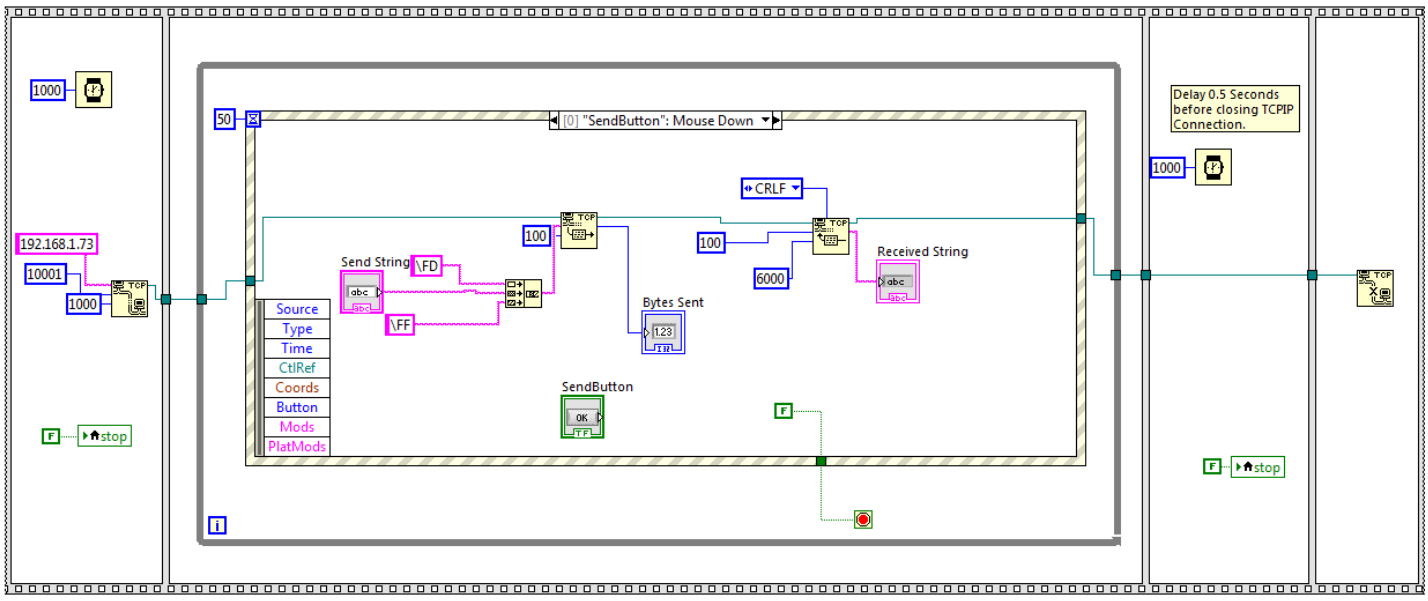



Figure 1: Block diagram of the LabVIEW program

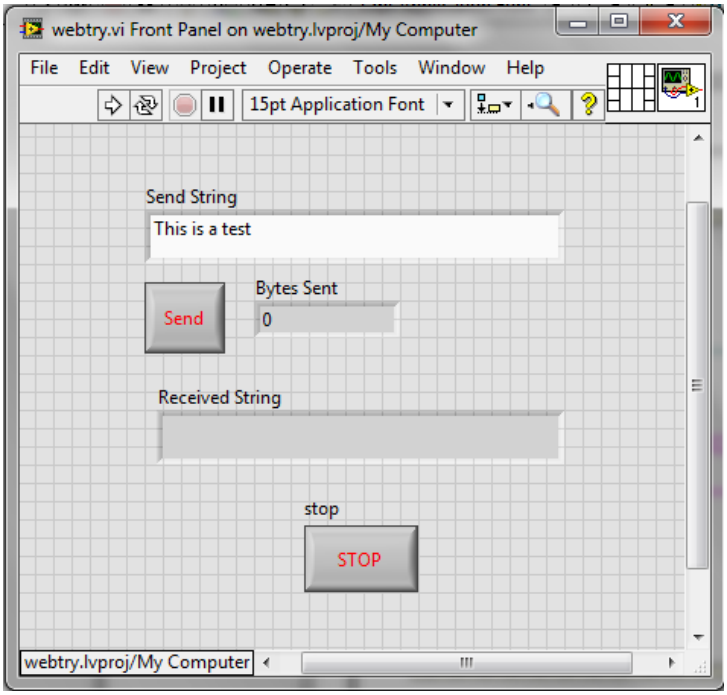



Figure 2: Front panel of the LabVIEW program

## Exercise 2

In this exercise we are back to working with Code Composer Studio and the F28379D Laundpad board (red board) on the robot car. Import "LABstarter", instructions in Lab 1 or your HowTo document, to create a new project in your workspace and call it `lab2<yourinitials>`. Once you have your new lab2 project perform the below steps.

1. For this lab, you will only be using CPU Timer 2's interrupt service routine "`cpu_timer2_isr(void)`". We will leave the timer0 and timer1 functions in our source code but we will not enable timer0 or timer1. So in `main()` find the two lines of code that set the TIE (Timer Interrupt Enable) bit to enable timer 0 and timer 1. Comment these two lines so they are not included in your program. i.e.

```
//CpuTimer0Regs.TCR.all = 0x4000;
//CpuTimer1Regs.TCR.all = 0x4000;
```

2. In `main()` find the "`ConfigCpuTimer`" function call for CPU Timer 2 and set its period to 0.25 seconds. Also find CPU Timer 2's interrupt function "`cpu_timer2_isr`." Note that in this function, it is blinking the blue LED on the Launchpad on and off. Build and Debug this code to make sure that the code compiles and runs. You should see the blue LED blinking on and off every half second. Once that is working, terminate your debug session so you are back in Edit mode. Before going to the next step, let's take a few minutes to think about the period value that you passed to the "`ConfgiCpuTimer`" function. 0.25 seconds is a large number of microseconds so you may have thought about what the largest acceptable number is to pass to this period parameter. To find this largest period setting we need to look at the TIM (timer register) and PRD (period register) registers of the CPU Timers. Both the PRD and TIM registers are 32 bits long and they each store a 32 bit unsigned integer. The TIM register starts at the value set in the PRD register and counts down by 1 every 1/200000000 seconds (200Mhz). (Note here that we are not use the pre-scaler of the CPU timer.) Whenever the TIM register counts down to 0 an interrupt event is issued calling the CpuTimer2 interrupt service routine. At this moment, the TIM register is also set back to the value set in the PRD register to start timing again.

   ☐ **TA Check-Off 3: Largest CPU Timer period**
   Knowing that a 32 bit unsigned integer has a maximum value, what is the largest period in seconds that the CPU Timers can be set to? **Explain your answer to your TA.**

3. Take a look at the global `uint32_t` variable `numTimer2calls`. Inside the `cpu_timer2_isr(void)` function, this variable increments by one each time that function is entered. In addition, every 10th time the function is entered, the global variable `UARTPrint` is set to 1. Change the code so that `UARTPrint` is set to 1 each time the `cpu_timer2_isr` function is entered. *You can do this by changing the modulus to 1 or better just remove the if statement.* Setting `UARTPrint` to 1, tells the `main()` while loop to print text through a UART serial port to the robot's text LCD dispaly. Find the `UART_printfLine(1, ...)` function call in the `main()` while loop.

   ☐ **TA Check-Off 4: Why UARTPrint is set and cleared**
   Does it make sense that when you set `UARTPrint` to 1, the while loop will call the `UART_printfLine (1, ...)` function? Why is `UARTPrint` set to zero inside the if statement after `UART_printfLine (1, ...)` is called? **Explain to your TA.**

4. Notice that the print statement prints the `numTimer2calls` global variable. Since `numTimer2calls` is a 32-bit integer the **%ld** formatter is used. Change the "line 2" `UART_printfLine(2, ...)` call so that both `4*numTimer2calls` and `6*numTimer2calls` is printed with a comma between the two numbers. Look at the line 2 call of `UART_printfLine` to see how to use two formatters to print two items.

> ☐ **TA Check-Off 5: Two numbers printed on the LCD**
> Debug and run your code and check that the LaunchPad's blue LED is still blinking and your text is printing to the LCD. **Show your TA.**

5. Write two worker functions "`void SetLEDsOnOff(int16_t LEDvalue)`" and "`int16_t ReadSwitches(void)`".

   - `void SetLEDsOnOff(int16_t LEDvalue)` takes a 16 bit integer as a parameter. The five least significant bits of this integer determine if the five LEDs are on or off. Bit 0 determines LED1's state. Bit 1 determines LED2's state. Bit 2 determines LED3's state. Bit 3 determines LED4's state. Bit 4 determines LED5's state. So for example if 18 (0x12, which is binary 10010) is passed to your function then LED5 and LED2 should be ON. Use five if statements inside your function to check, using the bitwise AND, &, operator, if the integer passed to your function has the least significant five bits either individually set or cleared. If set, turn ON the corresponding LED. If cleared, turn OFF the corresponding LED. See the above tables for LED GPIO assignments. I want you using the GP?SET and GP?CLEAR registers to turn on or off the LEDs. To test this function call it each time in your CPUTimer2 interrupt function and pass to it your global variable `numTimer2calls`. `numTimer2calls` is getting incremented by one each timer interrupt so you should see the five LEDs counting in binary.

     > ☐ **TA Check-Off 6: SetLEDsOnOff counting in binary**
     > What happens if the number passed to `SetLEDsOnOff()` is greater than 31? Also what happens when you pass a 32 bit integer to a function that is asking for a 16 bit integer like your `SetLEDsOnOff` function? **Demo and explain to your TA before writing your `ReadSwitches()` function below.**

   - `int16_t ReadSwitches(void)` returns a 16 bit integer that the least significant four bits indicates the state of the four push buttons. (Note that when each of the push buttons are not pressed the GPIO pin reads a 1 or high voltage. When pressed the GPIO pin reads a 0 or ground. This is because the IO pin is using an internal pullup resistor.) This function should have four if statements and use the bitwise OR, "|" operator to appropriately set bits of a local variable that will be returned by this function. So start the return variable at zero. Then if switch 1 is pressed OR 0x1 with local variable. If switch 2 is pressed OR 0x2 with variable. If switch 3 is pressed OR ??? with variable. If switch 4 is pressed OR ??? with variable. Finally return the local variable with the "return" instruction. See the above table for the GPIO pins that are connected to the push buttons and that are setup as inputs with pull-up resistor enabled in the default code.

6. Now that you have these worker functions, make your program a bit more interesting. Add code in your CPU timer 2 interrupt function so that you display to the LEDs the value returned from your `ReadSwitches()` function. Do this by creating a global `int16_t` variable and assign it the value returned from `ReadSwitches()`. Pass this global variable to your `SetLEDsOnOff(value)` function to see its binary value displayed on the LEDs. Also print this global variable by adding text (or shortening the text so it fits on the LCD) to one of the `UART_printfLine` function calls in `main()`'s while loop. Make sure to use the `%d` formatter because this is an `int16_t` variable.

   > ☐ **TA Check-Off 7: Push buttons shown on the LEDs**
   > **Show this working to your TA.**

## Exercise 3

1. Continue with your code from Exercise 2. Change the period of CPU timer 2 to 0.1 seconds.

2. Create another global `int32_t` variable similar to "`numTimer2calls`" but call this variable "`countsWhenNotPressed`". Then in your `cpu_timer2_isr`, for now, increment this new global variable by one each time `cpu_timer2_isr` is

called just like you did for the `numTimer2calls` variable. Pass this new count variable to the `SetLEDsOnOff()` function to display the least significant 5 bits of your count variable to the five LEDs. *This is similar to what you coded to test your `SetLEDsOnOff()` function in exercise 2.* Compile, download to the DSP and verify that indeed the LEDs are counting in binary.

3. Add one more item to this code as an exercise to see the use of bitwise operators in C. Calling the `ReadSwitches()` function, use an "if" statement and the bitwise C operator & to check if push buttons 2 and 3 are pressed. If both of these push buttons are pressed, stop incrementing the "`countsWhenNotPressed`" variable. If one or both of the pushbuttons are released, continue counting. Again, compile and download to the microcontroller.

   ☐ **TA Check-Off 8: Counter that stops with buttons 2 and 3**
   When your code is working, **demonstrate your application to your TA**.

## Exercise 4: Breakpoints and Watch Windows

Starting with the code you just finished, we want to experiment with adding breakpoints to your code and using the "Expressions window" to edit the values of your variables.

1. In your previous code (with the DSP halted), put your cursor over the integer variable that you are incrementing. You should see that the value of the variable appears. Run your code, halt it again, and again put your cursor over the variable to confirm that it changes.

2. An easier method than using the cursor repeatedly is to add the variable to the Expressions window. When the DSP is halted, the Expressions window displays the current value of each variable in the Expressions window. To add your counting integer variable to the Expressions window, highlight the variable and then right-click, then select **Add Watch Expression...**. The variable will appear in the Expressions window with the current value of the variable. The Expressions window dialog is also found under the View menu.

3. Next play a bit with adding breakpoints and single stepping through a section of code. The code you have written to this point is very small. Add the following nonsense code to allow for easy use of breakpoints and code stepping. At the top of your C-file, but below the #includes, add the following global variables:

```
float x1= 6.0;
float x2= 2.3;
float x3= 7.3;
float x4= 7.1;
```

   Then inside your CPU timer 2 interrupt function add this nonsense code:

```
x4 = x3 + 2.0;
x3 = x4 + 1.3;
x1 = 9*x2;
x2 = 34*x3;
```

   Build and load your code. Add a breakpoint to your code by double clicking on the left gray margin of your source file. A breakpoint is a location where the program will literally halt during execution. This allows you to check the values of your variables during operation. After a breakpoint, you can single step through your code (F5) and watch the variables update as different calculations are performed. You remove breakpoints by again clicking in the left gray margin.

4. If you happened not to receive any compiler errors during any of the above exercises, you should intentionally add some errors to your code so that you will see how CCS will alert you during the build process. Try double clicking on the error message. The editor will then take you to the line of code that has the error.

> ☐ **TA Check-Off 9: Breakpoints and the Watch Window**
> Demonstrate that you know how to use Breakpoints and the Watch Window to debug your source code.

## Exercise 5

For many of our lab assignments we will want to have at least one of our timers running at a fast periodic rate. Most of the time that will be somewhere between a period of 1ms to 5ms. I would not be surprised though, if some of your projects will require you to run code at an even faster rates and the F28379D can definitely handle periodic rate of 0.1ms to 0.02ms. For this exercise, use a period of 1ms. which can also be stated as a sample frequency of 1Khz. Change CPU Timer 2's period to 1ms in order that CPU Timer 2's interrupt function is call once every millisecond.

The F28379D can perform a large number of instructions every 1ms, but there are some things you do not want the processor performing every 1ms. For example the printing to the robot's text LCD, if we printed every 1ms our eyes would not be able to see all the text spilling to the screen. Also calling the `SetLEDsOnOff()` function every 1ms would cause a blur if LED changes. Add code to your CPU Timer 2 interrupt function to only print every 100$^{th}$ time the function is called. The % (mod) operator is perfect for this. Mod returns the remainder of an integer divided by another integer. i.e. (56 % 5) = 1. Using "`numTimer2calls`" that you are incrementing every time in the timer interrupt, write an if statement with a % (mod) condition that causes the if statement to be true every 100$^{th}$ time in the timer interrupt. Inside this if statement, perform all code that makes sense to run at the slower rate.

> ☐ **TA Check-Off 10: 1 ms timer period code**
> **Demo this to your TA.**

**Now that you are done with the lab power off your Raspberry Pi by** going back to the Putty terminal that is connected to your PI and typing "`sudo shutdown now`". This will shutdown the Pi and it takes about 5 seconds or so to shutdown.

## Lab Check Off

1. Demonstrate your LabVIEW programming talking back and forth to your robot's Raspberry Pi board.
2. Demonstrate your first application that continually checks the status of the four pushbuttons and displays their current state on the five LEDs. One LED should always be off since there are only four push buttons.
3. Demonstrate your second application that updates a counter every quarter second and outputs the least significant 5 bits of the count to the five LEDs. The count should stop if both pushbuttons 2 and 3 are pressed and resume when one or both of them are released.
4. Demonstrate that you know how to use Breakpoints and the Watch Window to debug your source code.
5. Demonstrate your 1ms timer period code working.
6. For your lab submission submit your working **commented** code to Gradescope. Take time to add comments explaining what you understand is happening in the code you wrote and the functions in which your code is running. Please make it obvious in your submission which code is for each exercise. I do not want short, hard to understand, comments. Instead, I would like short paragraphs explaining the code you wrote.

## Check-Off Summary

Tick each box once your TA or instructor has checked you off.

- [ ] Check-Off 1: Meaning of the if statement
- [ ] Check-Off 2: LabVIEW talking to the Raspberry Pi
- [ ] Check-Off 3: Largest CPU Timer period
- [ ] Check-Off 4: Why UARTPrint is set and cleared
- [ ] Check-Off 5: Two numbers printed on the LCD
- [ ] Check-Off 6: SetLEDsOnOff counting in binary
- [ ] Check-Off 7: Push buttons shown on the LEDs
- [ ] Check-Off 8: Counter that stops with buttons 2 and 3
- [ ] Check-Off 9: Breakpoints and the Watch Window
- [ ] Check-Off 10: 1 ms timer period code

**Make sure that you comment you code and put your initials in your Gradescope submission! See https://marius-juston.github.io/SE-423---Class-Material/Homeworks/commenting_code_example.pdf for how we expect the code to be commented like!**